

Closed for the Weekend

Prediction Markets as an Upstream Layer of Price Discovery

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Abstract

Each Friday night the US equity market goes dark; the events that determine firm values, however, never pause. I exploit this recurring institutional window — Friday 8:00 p.m. through Sunday 8:00 p.m. Eastern — in which US listed equities cannot trade on any venue, while prediction markets on Polymarket and Kalshi remain open. Using 749 prediction market to stock links, of which 501 are assigned an explicit economic sign, I ask whether probability revisions formed inside this blackout period can predict the linked stock's peer-adjusted reopening value, and the answer is a resounding yes: a 10 percentage points favorable event market revision predicts a 27.7 basis points favorable peer-adjusted reopening move within the same repeatedly observed link. A portfolio of exposed stocks sorted on signal strength earns a high-minus-low spread of 37.8 basis points. The driving component is not raw activity: the stock response is stronger when Kalshi and Polymarket agree and when movement is shared across sibling contracts, while venue-specific and leader-specific residuals, as well as prediction-market notional and trade counts are uninformative. Larger absolute prediction markets revisions are followed by wider opening price swings and higher realized volatility but by no abnormal equity volume. Aggregating several event exposures is more informative than selecting one leading contract, suggesting that firms can be represented as portfolios of dynamically priced event states. The results support a view of event markets not merely as forecasting tools, but as an upstream information layer that can make real-world states observable before conventional securities markets are able to respond.

Keywords: prediction markets, price discovery, market closures, information aggregation

1 Introduction

Markets close; the world does not. Elections are held, courts rule, legislators enact and regulators announce, wars escalate and de-escalate, firms are innovating, and intentions are made clear in political speeches. Clearly some of that news can be relevant for the value of a specific listed company. Under the classical view of market closures that dates to [French and Roll \(1986\)](#), the relevant information accumulates in a latent state and reveals itself through returns and variance at the reopening — accumulation itself is not visible, only the consequence thereof. More broadly, empirical work has made otherwise latent information and expectations observable using text-based measures, surveys, and prices of related financial claims ([Baker et al., 2016](#); [Atanasov et al., 2017](#); [Bernanke and Kuttner, 2005](#)). Historical natural experiments can isolate periods in which information could or could not arrive ([Koudijs, 2016](#)), and the after-hours and preopening research shows that meaningful price discovery can occur outside of ordinary trading ([Barclay and Hendershott, 2003](#); [Biais et al., 1999](#); [Cao et al., 2000](#); [Madhavan and Panchapagesan, 2000](#)). Yet the central object remains difficult to observe: what did the market learn about a firm while its equity could not trade?

The two largest prediction-market venues — Kalshi and Polymarket — continuously trade thousands of contracts quoting precisely the kinds of political, legal, regulatory, and geopolitical events that can move firm values. In recent years their contracts have become broad enough that one can now match a meaningful cross-section of contracts to individual listed firms. As a consequence, during the weekend equity market closure, information about a firm can be priced in a market where the firm is not even listed. A contract may ask whether a candidate wins an election, a certain geopolitical event occurs, or a product crosses a measurable threshold. Its price is therefore a traded summary of beliefs about an event that can affect cash flows, discount rates, competitive conditions, or policy exposure of a company. Moreover, prediction-market prices don't have to be interpreted as literal objective probabilities to be informative state variables ([Wolfers and Zitzewitz, 2004](#); [Grossman and Stiglitz, 1980](#)). The economically relevant question is whether changes in those state variables map into the values of exposed securities.

The present paper studies that mapping in a setting that imposes temporal order. I define a hard closure as a period during which US listed stocks are mechanically unable to trade, and measure the prediction market (PM) signal accumulated from Friday 8:00 p.m. to Sunday 8:00 p.m. Eastern. It ends on Sunday evening, and not with the Monday opening bell, because overnight trading on alternative venues for some securities can resume as early as 8:00 p.m. Sunday, and after that moment one can no longer assert that the linked equity was mechanically unable to move and potentially leak back into the prediction market price. Ending the window there costs some statistical power but solves the simultaneity problem during signal formation, because the linked equity cannot trade inside the measured window.

The main data used are minute-level prediction-market trades and price paths from Polymarket and Kalshi and minute-level linked stocks' quotes and trades sourced from Databento. Constructing the economic links requires an additional mapping step. To reduce roughly 100,000 largest prediction-market contracts to a tractable set of economically plausible firm exposures, I use three frontier large language models as independent first-pass screeners.

Each model proposed prediction-market to stock links separately, was then shown the other models' suggestions, and iteratively critiqued and revised its own candidates until the three sets converged. This consensus procedure reduced the universe to roughly 2,000 candidate event-stock relationships, which were subsequently reviewed manually for economic validity. Each accepted link was assigned a direction indicating whether a higher probability of the contract's YES outcome should benefit or harm the linked firm. This produced 749 reviewed links, of which 501 satisfy the data and directional requirements used in the main signed analysis. Because the mapping is necessarily judgment-based, I separately validate the proposed links and signs using ordinary trading periods that are excluded from the weekend experiment.

Finally, the equity reopening outcome is constructed from two independent data channels rather than relying on a single opening quote or trade. Quote- and transaction-based early-session prices are retained only when they corroborate one another under an outcome-blind agreement rule calibrated outside the analysis sample. The resulting reopening gap is benchmarked against matched same-date firms that have no exposure to the underlying event.

The first result establishes a within-link conversion rate between event probabilities and firm value. Using repeated observations of the same prediction-market to stock relationship, I find that a one-percentage-point favorable revision in the prediction market is associated with a 2.77 basis-point favorable peer-adjusted stock reopening move. Because identification comes from variation within the same economic link, the estimate is not driven by permanently more important firms or contracts. I then test whether this result depends on the regression-imposed linear functional form. On each reopening date, I sort stocks into high- and low-signal groups according to their signed prediction-market revisions. On average the high-signal portfolio outperforms the low-signal portfolio by 37.8 basis points. This confirms that prediction-market information ranks exposed firms cross-sectionally even without assuming a constant marginal response to each additional probability point.

Those estimates establish that the event-market state contains equity-relevant information. They do not yet reveal what kind of information it is. The paper's second contribution is to identify what I call the corroboration premium. In stock-weekends with valid signals on both Kalshi and Polymarket, the response is sharply stronger when the two platform-level signals point in the same economically signed direction. The cross platform test does not require identical contracts to be listed on both venues: it compares the largest valid platform-level signal for the same stock on the same weekend. A separate, stricter event-group disagreement analysis finds no stable platform winner. The result suggests that independently reproduced movement is more informative than isolated movement.

A second corroboration result appears within families of related contracts. Decomposing a leading contract's revision into a component shared by sibling contracts and a leader-specific residual, the common component predicts the stock reopening move with a coefficient of 6.662 ($p = 0.0084$), while the residual is essentially zero (-0.166 , $p = 0.838$), and the difference between the common and residual components is itself significant. Taken together, the cross-platform and sibling-contract results show that the common component of prediction-market

movement carries the predictive content, while the idiosyncratic component does not.

Next, the paper moves from a view of single event exposure to a firm-level portfolio of event exposures. When several event markets are linked to the same stock, the capped sum of economically signed revisions has a standardized coefficient of 0.0794 ($p = 0.0019$), while the event-family capped sum is similarly strong at 0.0806 ($p = 0.0012$). Other aggregation rules point in the same direction: the median signed revision yields a coefficient of 0.0702 ($p = 0.0301$), the simple mean 0.0694 ($p = 0.0445$), the notional-weighted mean 0.0682 ($p = 0.0514$), and the top-three capped sum 0.0579 ($p = 0.1058$). By contrast, selecting only the single largest-moving contract produces the weakest estimate, 0.0315, and is statistically insignificant ($p = 0.3607$). I interpret this pattern as suggestive of a potentially new way of representing a firm as a portfolio of dynamically priced event exposures: a firm is simultaneously exposed to political, regulatory, legal, product, geopolitical, and other contingencies, and their combination is more informative than any individual contract alone.

Finally, I find evidence that price discovery often takes place “silently”. The absolute PM revision is positively associated with the stock’s first-five-minute price range and first-thirty-minute realized volatility, but not with relative volume over either of the windows. Raw PM notional and trade counts also have no effect. The evidence does not say that “busy” prediction markets predict busy stocks. It says that a directionally interpretable and corroborated state revision maps into the linked stock price even when trading volume does not grow. This separates information quality from simple attention.

The timing analysis asks when equities absorb information already reflected in prediction markets. The signal is visible in transaction prices by 9:25 a.m., with evidence of more adjustment during the final premarket interval. After the regular session begins, however, there is no reliable continuation in the prediction market implied direction beyond the first minute and no subsequent reversal. The evidence therefore places substantial incorporation by late premarket trading and the opening process, with little additional predictable adjustment once regular trading is underway.

This paper sits at the intersection of several established literature traditions. First, it builds on the market-closure literature beginning with [French and Roll \(1986\)](#), which shows that information accumulates while securities are unable to trade and is subsequently revealed through reopening returns and volatility. The present setting adds an observable object to that framework: prediction-market prices evolve during the closure and therefore provide a direct measure of part of the information accumulated while equities are silent. Second, the paper relates to the prediction-market and market-based expectations literature. Relative to classic work, which emphasizes forecasting and information aggregation ([Wolfers and Zitzewitz, 2004](#); [Forsythe et al., 1992](#); [Arrow et al., 2008](#)), I ask whether event probabilities have external valuation content for another asset market. Relative to market-based event-study identification ([Snowberg et al., 2007](#); [Bernanke and Kuttner, 2005](#)), I study many event classes and many firm exposures rather than one policy or election episode, and exploit a window in which the linked stocks cannot contemporaneously respond. Third, the paper connects to cross-market price-discovery research, but differs from studies of stocks, options, and fragmented exchanges because the prediction markets trade events that a firm could

be exposed to rather than another financial claim on the firm. Finally, the paper builds on research on economic links and political risk by mapping event probabilities into firm exposures. This creates a continuously observed, market-based representation of firm-specific event risks and suggests that firms can be viewed as portfolios of dynamically priced event states.

The broader contribution is an information architecture. Traditional price discovery is usually modeled within a hierarchy of financial claims: stocks, options, futures, bonds traded across multiple exchanges – all linked to the same underlying asset. Event markets introduce a different upstream layer. They translate states of the world into traded probabilities. During a hard closure, those probabilities can move before the underlying security can. Once the security reopens, the information migrates into conventional prices. Event probabilities can make otherwise latent state expectations observable as continuous variables, and portfolios of such states’ exposures become a new representation of a firm while its stock is silent.

The rest of the paper proceeds as follows. Section 2 places the study in the literature. Section 3 describes the institutional setting and develops the identification strategy. Section 4 presents the prediction-market and equity data, the economic mapping between event contracts and firms, and the construction and validation of the main signal and reopening outcome. Section 5 presents the main evidence on the transmission of prediction-market probability revisions into linked equity values. Section 6 examines the corroboration premium across independent prediction-market platforms and related contracts. Section 7 extends the analysis from individual contracts to firms as portfolios of event exposures. Section 8 studies the distinction between price adjustment and trading intensity. Section 9 traces when the prediction-market signal becomes incorporated into equity prices. Section 10 discusses the broader implications for price discovery, expectations measurement, and event-based representations of firm risk. Section 11 concludes.

2 Related literature

2.1 Market closures

The modern literature on market closures begins with French and Roll (1986), who compare return variance across trading and nontrading periods and use unusual exchange closures to distinguish public-information arrival, private information incorporated through trading, and trading-related noise. Their key point is that periods in which markets cannot trade are informative in their own right. Barclay et al. (1990) use overnight and trading-period return variances to study when private information enters prices, while Barclay and Hendershott (2003) show that meaningful price discovery can occur after hours even when trading volume is relatively low. Barclay and Hendershott (2004) further document the adverse-selection costs associated with that trading. Koudijs (2015, 2016) obtain especially clean identification from eighteenth-century sailing schedules, where the arrival of a boat determined when information could reach Amsterdam. My setting differs in one important respect: part of the information accumulated while the equity is closed is directly observed through prediction-market prices.

The contribution is therefore not to show that information arrives during market closures, but to measure part of that information before it appears in the stock price.

Because the paper focuses on the reopening price, the microstructure of the opening matters. [Amihud and Mendelson \(1987\)](#) show that returns measured at the opening differ systematically from returns generated during continuous trading, while [Stoll and Whaley \(1990\)](#) relate opening volatility to market structure. [Biais et al. \(1999\)](#) show that the preopening period in the Paris Bourse is a genuine learning process rather than cheap talk, [Cao et al. \(2000\)](#) document price discovery in the Nasdaq preopening in the absence of trading, and [Madhavan and Panchapagesan \(2000\)](#) study how the specialist’s price aggregates order flow. [Gerety and Mulherin \(1992\)](#) examine trading volume around market openings and closings.

This literature also matters for measurement. Noisy or asynchronous prices can create misleading return relations. Bid–ask bounce and transaction-price noise can bias measured returns ([Blume and Stambaugh, 1983](#)), and noisy prices can affect inference rather than simply add variance ([Asparouhova et al., 2010, 2013](#)). These concerns motivate two choices in the paper. First, I construct the reopening outcome from both quote and trade data rather than rely on a single opening price design. Second, I treat the opening as a process rather than a single timestamp, which motivates the use of horizon-specific quote, trade, and consensus prices in the timing analysis.

2.2 Prediction markets and market-based expectations

Prediction markets aggregate dispersed beliefs about future events into prices of event-contingent claims ([Wolfers and Zitzewitz, 2004](#); [Arrow et al., 2008](#)). The basic idea goes back to [Hayek \(1945\)](#): market prices can summarize information that is dispersed across many individuals. [Grossman and Stiglitz \(1980\)](#) make the related point that prices cannot reveal all information perfectly when acquiring information is costly. For the purposes of this paper, prediction market prices do not need to be interpreted as perfectly calibrated objective probabilities. What matters is that changes in those prices contain information about the underlying event and, in turn, about firms that are economically exposed to it.

A large literature studies how well prediction markets perform in that role. [Forsythe et al. \(1992\)](#) show that a small real-money political market could outperform contemporaneous polling, while [Wolfers and Zitzewitz \(2004\)](#) survey the broader evidence on prediction-market accuracy and information aggregation. [Arrow et al. \(2008\)](#) discuss the case for wider institutional use of prediction markets. On the theory side, [Ostrovsky \(2012\)](#) derives conditions under which dynamic markets with strategic traders aggregate information, and [Atanasov et al. \(2017\)](#) compare prediction markets with structured prediction polls. Most of this literature evaluates prediction markets against the outcomes of the contracts themselves.

The paper closest to the question studied here is [Snowberg et al. \(2007\)](#). They use election prediction-market probabilities to measure changes in the likelihood of political outcomes and relate those changes to contemporaneous movements in financial markets. Their approach shows how a traded expectation can turn a discrete political event into a continuously measured state variable. A similar measurement logic appears in [Bernanke](#)

and Kuttner (2005), who use federal funds futures to separate expected from unexpected monetary-policy decisions. I use the same broad idea in a different setting. Rather than focusing on a single election or policy decision, I study prediction-market contracts covering political, regulatory, legal, geopolitical, product, and corporate events and map those event probabilities to individual firms.

2.3 Cross-market price discovery

Another relevant strain of literature asks where price discovery occurs when the same security, or closely related claims on it, trade in several places. Hasbrouck (1995) provides the standard framework for this question. When the same security trades on several venues, the observed prices are tied together by a common long-run value even though they may differ temporarily. Hasbrouck models these prices jointly in a vector error-correction system and decomposes innovations into temporary movements and changes in the common permanent price component. A venue’s information share is the fraction of variance in that permanent innovation attributable to shocks that first appear in that venue. His later work applies the same logic to price discovery across equity-index products (Hasbrouck, 2003).

A closely related question arises across stocks and their derivatives. Easley et al. (1998) argue that informed traders may choose to trade in options and show that option-market activity can contain information about subsequent stock prices. Chakravarty et al. (2004) likewise find an important role for options in price discovery. Muravyev et al. (2013), however, show that much of the apparent informational advantage of option quotes disappears once stock and option quotes are aligned carefully in time and microstructure effects are addressed. This debate is particularly relevant here because apparent lead–lag relations can be created by differences in trading hours, quote timing, or noisy prices rather than by genuine informational leadership. Related work studies how price discovery is affected by fragmentation and order routing across equity venues (Foucault and Menkveld, 2008; O’Hara and Ye, 2011).

My setting differs from these studies in two important ways. First, the prediction-market contract and the stock are not alternative claims on the same payoff. A prediction-market contract pays on a particular event, while the stock reflects the value of the entire firm. There is therefore no reason for their prices to share a common permanent component in the sense required by the Hasbrouck framework. Their connection is economic: the firm is exposed to the event being priced. Second, the two markets do not trade simultaneously during the signal window. Between Friday 8:00 p.m. and Sunday 8:00 p.m., the prediction market can update while the linked equity cannot trade. A conventional information-share decomposition is therefore neither necessary nor appropriate. Instead, the hard closure provides a direct test of whether an event-market revision formed while the stock is unavailable predicts the value at which the stock subsequently reopens.

2.4 Economic links and firm event exposures

The PM–stock mapping is closely related to the literature on economic links between firms. Cohen and Frazzini (2008) show that returns of major customers predict subsequent returns of

their suppliers, consistent with information being transmitted through economically important customer–supplier relationships. The connection in this paper is similar, except that the source of information is not another firm. It is a traded event whose outcome can affect the value of the linked company. As in [Cohen and Frazzini \(2008\)](#), the economic relationship is specified before examining the destination stock’s return: the question is whether information about an economically relevant source is subsequently reflected in the price of the exposed firm.

A related political-economy literature shows why the same event can have very different effects across firms. [Wagner et al. \(2018\)](#), for example, document substantial cross-sectional differences in stock-price responses to the 2016 US election depending on firms’ tax, trade, and regulatory exposures. [Hassan et al. \(2019\)](#) construct firm-level measures of political risk from earnings-call transcripts and show that political exposure varies considerably across firms rather than being a purely aggregate phenomenon. [Pástor and Veronesi \(2012, 2013\)](#) develop asset-pricing models in which uncertainty about government policy affects stock prices and risk premia, while [Kelly et al. \(2016\)](#) study the pricing of political uncertainty in option markets. Other event studies use sharp political shocks to value specific exposures: [Fisman \(2001\)](#) estimates the value of political connections from news about President Suharto’s health, and [Dube et al. \(2011\)](#) use pre-announcement trading around covert US coup authorizations to study the value of political ties.

Prediction markets provide a different way to measure these exposures because they attach continuously updating prices to specific events themselves. Once an event is linked to a firm and the direction of the exposure is specified, changes in the event probability can be related directly to changes in firm value. The multi-contract results push this idea one step further: a firm may be exposed to several political, regulatory, legal, product, or geopolitical events at the same time, so it can be useful to think of the firm as carrying a portfolio of event exposures rather than being represented by a single contract.

3 Institutional setting and identification

3.1 The hard closure

A US listed common stock can trade in four regimes: the regular session (9:30 a.m. to 4:00 p.m. Eastern), the post-close session (4:00 p.m. to 8:00 p.m.), for a subset of securities, an overnight session on alternative trading systems that opens Sunday at 8:00 p.m. and ends on Monday at 4:00 a.m., and the premarket session (from 4:00 a.m. to 9:30 a.m.). Between Friday 8:00 p.m. and Sunday 8:00 p.m. Eastern, none of these are available. I refer to this interval as the hard closure: the recurring window in which US equity cannot be traded on any venue.

Prediction markets follow no such rules. Polymarket and Kalshi operate continuously, and their contracts reference outcomes whose realizations directly affect the cash flows of listed firms. The hard closure therefore produces a strict informational hierarchy by the means of institutionally imposed barriers.

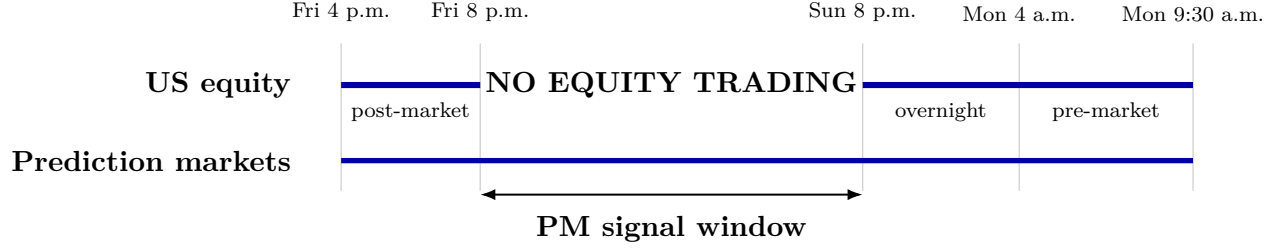


Figure 1. **The hard closure and the prediction-market signal window.** The prediction-market signal is measured from Friday 8:00 p.m. to Sunday 8:00 p.m. Eastern, when ordinary US listed equities cannot trade. Prediction markets remain active throughout the interval. Equity trading resumes only after the signal window ends, so contemporaneous stock-price feedback cannot take place in generating the measured prediction-market revision.

3.2 Identification

Studies that compare two simultaneously trading venues must confront the possibility that the apparent leader is in fact following: option quotes may respond to stock prices within milliseconds, and the statistical decompositions typically used to allocate price discovery (Hasbrouck, 1995) are attempts to resolve exactly this ambiguity. The hard closure removes the ambiguity by construction. During the measurement window there is no stock price to lead or follow. Whatever the event contract does between Friday and Sunday evening cannot be a response to contemporaneous equity trading in the linked firm, because no such trading exists.

The hard closure establishes temporal ordering, but it does not identify a causal effect of prediction-market trading on stock prices. The same public news can move the event contract over the weekend and the linked stock when trading resumes on Monday. The coefficient should therefore be interpreted as evidence of informational leadership: the prediction market reflects an economically relevant state before the equity is able to respond. The paper's claim is that this signal is observable and predictive of the subsequent stock price, not that Polymarket traders push stock prices around.

4 Data and measurement

4.1 Raw sources

The equity data come from Databento’s U.S. equities feed (EQUUS.MINI). I collect one-minute top-of-book best bid and offer quotes (bbo-1m) and one-minute open–high–low–close–volume trade bars (ohlcv-1m). The two series provide complementary views of the opening price. Quotes give a contemporaneous midpoint but can be wide, stale, one-sided, or crossed around the open. Trade bars are based on executed transactions, but the first observed trade can be small, venue-specific, affected by bid–ask bounce, or occur before the primary listing market has completed its opening process. The OHLCV open is simply the first qualifying trade recorded in that one-minute interval and is therefore not the same as the official primary-market opening-auction price. For this reason, the main reopening-price estimator uses both quote and trade information and retains observations only when the two sources agree sufficiently closely. I also collect SPY (SPDR S&P 500 ETF Trust) as a market benchmark.

The prediction-market data consist of minute-level price histories and full trade records for Polymarket and Kalshi, collected directly from the platforms’ public APIs. For Polymarket, I obtain market metadata and contract identifiers through the Gamma and CLOB APIs and retrieve one-minute probability histories from the CLOB price-history endpoint. Trade histories come from the Data API and, where needed to complete the record, from on-chain order fills in the Polymarket order-book subgraph. Because binary Polymarket contracts are represented by separate YES and NO tokens, I synchronize the two token histories and reconcile them into a single probability series. For Kalshi, I collect one-minute candlesticks and transaction records through the public Trade API. Kalshi provides recent active-market observations through live endpoints and older or settled-market observations through historical endpoints. **The accepted reopening sample spans February 26, 2024 through July 20, 2026. Coverage is unbalanced across venues and contracts because prediction-market listings begin and end at different dates; the empirical sample therefore uses the dates on which each reviewed link has valid prediction-market and equity data.**

4.2 The economic mapping

The central step is the mapping from event contracts to firms and, more importantly, the assignment of an economic direction for each pair. Candidate links were generated from metadata for the fifty thousand largest markets on each platform using three frontier large language models (ChatGPT Sol, Claude Fable, and Gemini Pro) as independent first-pass screeners, each proposing links, then critiquing and revising against the output of the other two until the proposals converged to a single candidate set of approximately two thousand events. Every candidate was then reviewed by hand and accepted or rejected as a legitimate economic relationship. That review produced 749 links covering 736 markets and 65 distinct stocks. Due to a number of technical failures — seventeen Polymarket markets that could not be collected, links with no Databento overlap, pre-listing and post-delisting issues, and

one invalid ticker mapping — 728 links remained eligible for joint analysis.

The unit of a link is a PM paired with a stock ticker. The mapping was built from direct firm, product, operational, regulatory, policy, demand, commodity, and other state variables and their implications for firm value. Because the prediction-market ecosystem expanded rapidly during the sample, the final panel is intrinsically unbalanced: markets begin and end at different times, firms have different numbers of linked contracts, and platform availability evolves. The specifications therefore rely on within-link variation and date fixed effects rather than treating the data universe as a balanced cross-section.

A raw probability movement is not economically meaningful until it is tied to a firm: a rising probability that a video platform is banned in the United States is unfavorable for that platform and potentially favorable for its competitors. Each link therefore carries an explicit sign $g_\ell \in \{+1, -1, 0\}$, where $+1$ means a higher YES probability benefits the linked firm, -1 means it harms the firm, and 0 means no monotone direction can be reasonably defended — for example, a contract on whether a CEO departs by the end of year, whose valuation depends on more nuanced and case-specific considerations. A zero does not mean the event is economically irrelevant. Among otherwise eligible links, 501 have $g_\ell \in \{-1, +1\}$ and enter the directional analysis.

Table 1 makes the mapping concrete. Panel A reports the exact reviewed, PM–equity data overlap, and signed composition by venue. Panel B shows a compact set of representative links selected from the reviewed mapping universe.

To make the construction concrete, consider Alphabet (GOOGL) on the April 20, 2026 reopening. The linked contract “Best AI in Dec 2026?” moved from 20.3% at the Friday boundary to 17.5% at the Sunday boundary. Its reviewed sign is $+1$, so the signed hard-closure revision is -2.8 percentage points. The first valid five-minute quote candidate is 338.375 and the five-minute trade candidate is 338.441, only 1.94 basis points apart; their consensus reopening value is 338.408. Relative to the previous close of 341.705, the raw reopening gap is -96.96 basis points and the SPY-adjusted gap is -80.97 basis points. The matched five-peer benchmark is -82.18 basis points, leaving a final linked-minus-peer reopening outcome of -14.78 basis points.

4.3 Mapping audit

The main empirical design requires two forms of economic judgment: whether a PM contract is meaningfully linked to a stock, and whether a higher YES probability is favorable or unfavorable for that stock. A natural question is: “why not let the data choose the links and signs?” A naive attempt to answer that question using the same weekend outcomes would be circular. If the stock-opening response were used to select signs, the main regression would mechanically inherit that choice. The validation sample therefore comes from periods that are absent from the hard-closure study. An even stronger objection is that both the links and the signs could have been chosen to produce the desired regression results, and the answer here is the same — audit the map on data that do not enter the experiment.

The validation panel excludes all 128 reopening dates, the 128 immediately preceding regular sessions, every Saturday and Sunday, and the entire Friday 8:00 p.m. to Sunday

Panel A: Mapping composition

Table 1. **Economic mapping and directional composition.** Panel A reports exact counts from the reviewed mapping. “PM–equity data overlap” requires usable prediction-market and equity data for the reviewed link over a common sample period. “Signed eligible” additionally requires a reviewed sign in $\{-1, +1\}$. The positive, negative, and nondirectional/unsigned columns partition the 728 links with PM–equity data overlap. Panel B gives representative examples of reviewed links.

Venue	Reviewed links	PM–equity data overlap	Signed eligible	Positive signed	Negative signed	Nondirectional / unsigned
Kalshi	132	129	97	68	29	32
Polymarket	617	599	404	305	99	195
All platforms	749	728	501	373	128	227

Panel B: Representative reviewed links

Venue	Contract	Stock	Sign	Economic rationale
Kalshi	Will a court order Google to be broken up before Jan. 1, 2026?	GOOGL	−1	Breakup reduces ecosystem value
Kalshi	Will SoFi be added to S&P 500 before Mar. 31, 2026?	SOFI	+1	Index inclusion raises index-fund demand
Polymarket	Will Meta be forced to sell Instagram or WhatsApp in 2025?	META	−1	Divestiture reduces ecosystem value
Polymarket	FDA approves Retatrutide this year?	LLY	+1	Approval raises pipeline value
Polymarket	Will Tesla (TSLA) beat quarterly earnings?	TSLA	+1	Beat signals stronger profitability
Polymarket	Will NVIDIA (NVDA) beat quarterly earnings?	NVDA	+1	Beat signals stronger profitability

8:00 p.m. hard-closure interval. The resulting sample contains 24,880 ordinary daily observations, 146,247 non-overlapping one-hour observations, and 24,437 same-date false-stock observations.

Using the fixed reviewed signs, economically favorable PM revisions are associated with favorable abnormal returns in the proposed firm outside the weekend experiment. The daily coefficient is 1.834 ($p = 0.0066$), and the one-hour coefficient is 0.597 ($p = 0.0066$). The result also survives persistent link-level sign randomization. In 9,999 draws, the reviewed daily orientation outperforms independent random signs ($p = 0.0006$), permutations preserving the observed overall positive/negative sign balance ($p = 0.0052$), and permutations preserving that balance separately within prediction-market platform ($p = 0.0054$).

Link validity is tested separately from sign validity. For every clean link-day, I compare the proposed stock with same-date false stocks. Without using any economic sign, larger absolute PM revisions predict greater absolute movement in the linked stock relative to the false-stock benchmark: the slope is 0.977 ($p = 0.0413$) across 24,437 observations and 701 links. Using the reviewed direction, the evidence is stronger: the PM revision predicts the linked stock’s return relative to the median false stock with $\beta = 1.625$ ($p = 0.0113$) across 17,656 observations and 482 links. Thus, the mapping does more than attach a plausible narrative to a moving market: the proposed stock tends to respond unusually strongly, and in the economically predicted direction, relative to contemporaneous counterfactual stocks.

Finally, I ask whether market data can independently recover the reviewed directions. Among 78 human-signed links for which repeated off-study variation is sufficient to estimate an empirical direction, the learned sign agrees with the reviewed sign in 47 cases, or 60.3%, against a 50% benchmark ($p = 0.0444$). In the stricter quality subset, 22 of 34 directions agree, or 64.7% ($p = 0.0607$). This exercise is intentionally asymmetric: strong historical evidence can corroborate or challenge a proposed direction, but failure to estimate a stable sign does not imply that the economic exposure is false. Whether a tariff decision helps or hurts a firm is a question about the firm’s supply chain, not a question that a short historical correlation can reliably answer. Many event exposures are sparse and state-dependent and may become relevant only when a particular political, regulatory, legal, or technological state moves sufficiently far.

To make this distinction concrete, Table 2 reports five cases in which an independently recovered economic direction disagrees with the reviewed economic sign. They illustrate why a human common sense based assignment need not coincide with a sign estimated from a finite history of market movement.

Table 2. **Examples of disagreement between reviewed and data-derived economic signs.** A reviewed sign of +1 means that a higher probability of YES was judged favorable for the stock; -1 means unfavorable. The final column gives the economic reasoning behind the reviewed sign.

PM contract	Stock	Reviewed sign	Data sign	Economic rationale for reviewed sign
Will Meta have the top AI model on June 30?	META	+1	-1	A stronger position in frontier AI ranking is favorable for Meta's competitive position and expected returns to AI investment.
Will egg prices be more than \$6.00 in March?	CALM	+1	-1	Higher egg prices are favorable for Cal-Maine because they drive producer's revenue and profitability.
U.S. Government funding lapse on March 15?	BAH	-1	+1	A federal funding lapse was coded as unfavorable for a major government contractor because shutdowns can delay government activity, contract execution, and associated revenue.
U.S.-China tariff agreement before 90-day deadline?	FIVE	+1	-1	A tariff agreement is coded as favorable for an import-exposed retailer because it may reduce its prospective merchandise costs.
New COVID variant of concern before 2027?	MRNA	+1	-1	A new medically important COVID variant was coded as favorable for Moderna through greater prospective demand for updated vaccines and renewed relevance of its COVID franchise.

4.4 The signed weekend signal

For reviewed link ℓ and reopening date d , define $p_{\ell,d}^F$ as the PM probability at the Friday 8:00 p.m. Eastern boundary and $p_{\ell,d}^S$ as the probability at the Sunday 8:00 p.m. boundary. Each boundary probability is the latest market probability observed at or before the stated timestamp. Then the signed hard-closure signal is

$$S_{\ell d} = g_{\ell} \times 100 \left(p_{\ell,d}^S - p_{\ell,d}^F \right), \quad (1)$$

where probabilities are represented on the unit interval and $S_{\ell d}$ is measured in percentage points. Positive $S_{\ell d}$ is therefore favorable to the linked stock by construction; negative $S_{\ell d}$ is unfavorable. Where a stock-weekend carries several eligible linked contracts, the primary link-level specification selects the contract with the largest absolute signed revision. Section 7 shows that this transparent choice is, if anything, conservative. The fixed clock window is central to identification. It is not chosen because all economically relevant news must arrive between those exact timestamps. It is chosen because it creates a clean period in which the destination equity is unable to trade.

4.5 The reopening price

The main outcome variable is the stock value established through the reopening price-discovery process. The natural candidate—the prevailing bid–ask midpoint at 9:30—is frequently unusable in precisely the securities this paper studies. Opening quotes in event-exposed stocks are often one-sided or extremely wide, and, critically, the pathology persists for several minutes rather than resolving within seconds. A return computed from such a quote can be not just noisy but entirely wrong. The obvious robust remedy — taking a median of the first few minutes of midpoints — does not help, because the contaminated minutes are exactly the ones being averaged. This is the concrete form of the general warning in [Blume and Stambaugh \(1983\)](#) and [Asparouhova et al. \(2010, 2013\)](#): noise in observed prices does not merely inflate variance, it biases the estimates and creates return relations that do not exist.

Trade prices have a different but equally important failure mode. The first observed transaction need not represent the value established by the primary opening process: it can be a very small print, occur on a peripheral venue, reflect bid–ask bounce, or precede completion of price discovery on the primary listing market. Moreover, Databento’s OHLCV bars aggregate qualifying trades in the interval rather than identifying an official opening-auction print, so an early bar can inherit an economically unrepresentative transaction without making that fact obvious. Averaging several bar closes reduces sensitivity to a single print but does not eliminate this problem and, by giving each minute equal weight, can itself overrepresent thinly traded intervals.

I therefore construct two candidate prices from independent data channels and accept an observation only when they corroborate each other. For stock s on reopening date d , let Q_{sd} be the first valid two-sided BBO midpoint observed in regular-session minutes 0–4 with

displayed relative spread no greater than 3 percent. Let T_{sd} be the volume-weighted typical trade price over the same five one-minute OHLCV bars, where the typical price in minute m is $(H_{sdm} + L_{sdm} + C_{sdm})/3$. The quote–trade disagreement is defined as

$$D_{sd} = \left| 10,000 \log \left(\frac{Q_{sd}}{T_{sd}} \right) \right| \quad (2)$$

The disagreement threshold is calibrated outside the paper’s analysis sessions. I remove every stock-date used anywhere in the analysis, and SPY, then split the remaining eligible sessions chronologically, using the first seventy percent for calibration and the last thirty percent for validation. The final calibration sample contains 5,157 usable quote–trade disagreements and the validation sample contains 5,230. The ninety-fifth percentile of the calibration distribution is $c = 125.17$ basis points. The later validation distribution is slightly tighter: 96.0% of validation disagreements fall below the frozen cutoff.

83.9% of accepted sessions obtain a usable quote in the 9:30 minute and 96.1% do so by 9:32, while the held-out disagreement distribution closely tracks the calibration distribution. When both Q_{sd} and T_{sd} exist and D_{sd} is below the calibrated threshold, the early-reopening consensus is

$$P_{sd}^{open} = \frac{Q_{sd} + T_{sd}}{2} \quad (3)$$

The design responds directly to a classic microstructure concern: noisy prices can create superficial return relations. By calibrating the quote–trade screen outside the main sample the outcome construction is designed to reduce contamination before the PM coefficient is estimated.

4.6 Market and peer adjustment

The accepted reopening value is compared with the previous session’s closing consensus and then benchmarked against matched same-date peers. Let $g_{s,d} = 10,000 \ln(P_{s,d}^{open}/P_{s,d-1}^{close})$ denote the raw reopening gap in basis points. The primary peer-adjusted outcome is

$$Y_{l,d} = g_{s,d} - \frac{1}{|\mathcal{P}_{l,d}|} \sum_{j \in \mathcal{P}_{l,d}} g_{j,d}, \quad (4)$$

where $\mathcal{P}_{l,d}$ is a set of matched same-date “false-link” peers carrying no economic exposure to the specific event under the reviewed map. Peers are selected using only pre-closure characteristics. For each principal stock-date, the procedure removes the ticker, firms linked to the event, and stocks that are themselves active prediction market linked observations on that date. It then ranks the remaining same-date candidates by size, trailing twenty-session gap volatility, previous-session intraday return, and previous-session final-thirty-minute return. The five nearest peers are retained whenever available, with at least three required. The median peer count is 5 with mean of 4.87. Appendix Table A1 reports matching balance and shows that main specification result is stable after directly controlling for focal-minus-peer characteristic differences and after weighting the same frozen peers by match distance.

4.7 Sample construction

Table 3 summarizes the final analysis layers. The first three rows are link-level mapping counts; the remaining rows are stock-weekend or date-level analysis samples. The important transition is the final quote–trade consensus requirement: 1,312 stock-weekends have an accepted primary reopening outcome, of which 1,173 enter the repeated-link specification in Section 5 after requiring complete predetermined controls and within-link identifying variation.

Table 3. **Sample construction and descriptive statistics.** Panel A reports the main mapping and estimation samples. Panel B summarizes the 1,312 stock-weekends with an accepted reopening outcome. Probability revisions are in percentage points; price and disagreement measures are in basis points.

Panel A. Final sample layers				
Stage	Count	Stocks	Dates	Markets
Reviewed PM–stock links	749	65	—	736
PM–equity data overlap	728	64	—	715
Signed eligible links	501	51	—	500
Accepted reopening sample	1,312	45	95	253
Primary within-link estimating sample	1,173	41	91	169
Multi-contract common support	1,278	43	91	250
Same-date portfolio sample	—	—	88	—
Panel B. Descriptive statistics for the accepted reopening sample				
Variable	Mean	SD	Median	P_{10} / P_{90}
Signed weekend revision S (pp)	−0.27	7.60	0.00	−7.00 / 5.97
Absolute weekend revision $ S $ (pp)	4.05	6.43	2.00	0.00 / 10.00
Peer-adjusted reopening gap (bps)	8.28	222.76	0.77	−217.69 / 227.18
Raw reopening gap (bps)	3.43	239.30	7.64	−250.06 / 226.13
SPY-adjusted reopening gap (bps)	−3.19	213.75	−4.89	−206.53 / 194.94
Quote–trade disagreement (bps)	34.59	30.10	24.56	4.34 / 82.61
Linked contracts per stock-weekend	3.42	3.78	2.00	1.00 / 8.00
Friday prior variance $p(1 - p)$	0.128	0.085	0.131	0.009 / 0.240

5 Event probability to firm value

The primary paper specification asks whether the signed weekend prediction market change predicts the peer-adjusted reopening value inside the same repeatedly observed economic relationship:

$$Y_{l,d} = \beta S_{l,d} + \alpha_l + \lambda_d + X'_{s,d-1}\gamma + \varepsilon_{l,d}. \quad (5)$$

The link fixed effects α_l do essential work. Different contracts can carry potentially opposing or incomparable economic weights for the same firm: a contract on a merger approval and a contract on a distant geopolitical outcome are not the same objects, and neither are the firms linked to them. The α_l coefficient removes everything time-invariant about contract type, exposure strength, firm characteristics, and average stock behavior, so that identification comes from a single comparison: larger versus smaller weekend revisions within the same reviewed relationship. The reopening-date fixed effects λ_d absorb whatever is common to a given Monday morning — macroeconomic news, market direction, the aggregate state. In the baseline specification, $X_{s,d-1}$ contains three measures of the stock’s recent return dynamics: the previous session’s intraday return, which absorbs short-run continuation or reversal in the stock’s most recent trading day; the return over the final 30 minutes of that session, which controls for late-day price pressure immediately before the closure; and the trailing 20-session volatility of opening gaps, which captures the stock’s pre-existing tendency to experience large or noisy overnight repricing. Expanded specifications additionally control for the previous abnormal opening gap, the trailing five- and twenty-session mean opening gaps, and five- and twenty-session momentum. These variables absorb persistent gap behavior, short- and medium-horizon reversal, and momentum.

Table 4 makes the specification progression explicit. Columns (1)–(6) deliberately keep the same 1,173 observations while moving from a minimal reopening-date fixed-effect specification to stock fixed effects, then to the reviewed PM–stock-link fixed effects that define the main within-link design, and finally add the three predetermined controls one at a time. This fixed-support sequence separates changes in the estimate caused by the specification from changes caused by sample composition. Column (7) then asks a distinct robustness question by adding the broader gap and momentum controls on their available common support.

The progression in Table 4 is economically stable. With reopening-date fixed effects alone, the slope is 2.219 basis points per probability point; adding stock fixed effects yields 1.878, and replacing them with the more demanding PM–stock-link fixed effects raises the estimate to 2.637. Sequentially adding the three predetermined controls leaves the coefficient essentially unchanged, ending at $\hat{\beta} = 2.773$ in the primary specification. Thus the main estimate is not created by a particular fixed-effect choice or by controlling for recent return and gap dynamics. A ten-percentage-point favorable weekend revision corresponds to approximately 27.7 basis points of favorable peer-adjusted reopening movement within the same reviewed link. The broader control set gives the same conclusion. Column (7) retains 1,129 observations and yields a coefficient of 3.129 basis points per probability point ($p = 0.0102$), so controlling additionally for persistent opening-gap behavior and recent momentum does not attenuate the relation.

Inference is potentially complicated because the observations are not independent. The same stock can appear on many weekends, the same prediction market can be linked to several observations, and all stocks reopening on a given date are exposed to the same weekend news and Monday market conditions. Standard errors that ignore these sources of dependence would therefore be too optimistic. Following [Abadie et al. \(2023\)](#) and [Cameron et al. \(2011\)](#), I report cluster-robust inference under alternative assumptions about where this dependence is

Table 4. **Specification progression for the within-link reopening outcome.** Columns (1)–(6) use the same main specification 1,173-observation support. Column (7) adds the expanded gap and momentum controls on their available common support.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Coefficient estimates							
Signed weekend revision	2.219** (0.883)	1.878** (0.955)	2.637** (1.244)	2.669** (1.277)	2.787** (1.271)	2.773** (1.264)	3.129** (1.217)
Prior-session intraday return	—	—	—	−0.043 (0.026)	−0.009 (0.022)	−0.009 (0.022)	0.029 (0.029)
Prior-session final-30m return	—	—	—	—	−0.212*** (0.068)	−0.210*** (0.068)	−0.214*** (0.074)
Trailing 20-session gap volatility	—	—	—	—	—	−0.017 (0.020)	−0.015 (0.043)
Panel B: Specification details							
Cross-sectional fixed effects	None	Stock	Link	Link	Link	Link	Link
Reopening-date fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Expanded gap and momentum controls	No	No	No	No	No	No	Yes
Observations	1,173	1,173	1,173	1,173	1,173	1,173	1,129

Notes: The dependent variable is the peer-adjusted reopening gap in basis points. Coefficients on S are basis points of reopening movement per one-percentage-point favorable prediction-market revision. Standard errors in parentheses are two-way clustered by stock and reopening date. Columns (1)–(6) deliberately hold the sample fixed while moving from date fixed effects to stock fixed effects, then reviewed PM–stock–link fixed effects, and finally the three predetermined controls in the primary regression. Column (7) additionally controls for the previous abnormal opening gap, trailing five- and twenty-session mean opening gaps, and five- and twenty-session momentum; it contains 167 links, 39 stocks, and 88 reopening dates. PM–stock–link fixed effects absorb all time-invariant stock and link characteristics. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

strongest, including two-way clustering by stock and reopening date, and two-way clustering by prediction market and reopening date.

I also use two procedures that place less weight on large-sample approximations. This matters because the sample contains only 91 reopening dates. The date-cluster wild bootstrap asks how unusual the estimated coefficient is when date-level shocks are repeatedly reweighted, while the within-date permutation test asks whether the observed pairing between prediction-market signals and stock outcomes is stronger than would arise if signals were reassigned across stocks on the same reopening date. The coefficient remains statistically significant under all four procedures. The wild-bootstrap and permutation p -values are 0.0070 and 0.0060, respectively.

Influence diagnostics tell the same story from a different angle. Every leave-one-stock estimate remains positive; every leave-one-date estimate remains positive; every leave-one-link estimate remains positive. Statistical significance varies as information is removed, but no single firm, opening date, or link flips the economic direction. Given that the sample period contains elections, wars, and high-profile rulings — episodes that could possibly carry a result single-handedly — this evidence is worth emphasizing.

Appendix Table A1 checks whether the main result depends on how the matched peer benchmark is constructed. The peers are chosen to be similar to the focal stock on predetermined characteristics, but the match is not exact. I therefore first add the remaining focal-minus-peer differences in size, recent gap volatility, previous-session return, and late-day return directly to the main regression. The estimated slope changes only modestly, from 2.773 to 2.637 basis points per probability point.

I then ask whether the result depends on giving all matched peers equal weight. Instead, I give greater weight to peers that are closer to the focal stock according to the same predetermined matching variables. Under this inverse-distance weighting, the slope is 2.812. These checks show that the main estimate is not driven by small remaining differences between focal stocks and their matched peers, or by the decision to weight all selected peers equally.

A regression slope can be influenced by a small number of large revisions; also linear regression imposes the assumption that every additional probability point carries the same marginal value. The next test discards both of these features. On each reopening date, eligible stocks are sorted into terciles by the signed weekend revision, and the equal-weighted high-minus-low peer-adjusted reopening portfolio is

$$\text{HML}_d = \bar{Y}_d^{\text{High}} - \bar{Y}_d^{\text{Low}}. \quad (6)$$

Since both averages are formed on the same morning, all broad weekend and Monday-morning conditions are held fixed by construction rather than by controls.

Across 88 reopening dates, the average high-minus-low spread is 37.77 basis points, with a median of 26.57 basis points. The result is statistically significant using both HAC inference, which allows for serial dependence across reopening dates ($p = 0.0042$), and a within-date assignment-randomization test that compares the observed sorting to random same-date assignments ($p = 0.0060$). These findings imply that an observer who knew only the direction and relative strength of the weekend event signals could rank exposed firms and separate

them by roughly a third of a percent with no need to assume any specific functional form.

Requiring at least ten stocks on a date leaves 34.54 basis points ($p = 0.0133$); dropping the single date with the largest absolute spread raises the mean to 43.24 ($p = 0.0003$); dropping the two largest leaves 38.69 ($p = 0.0005$); and winsorizing date-level spreads at 2.5% leaves 39.83 ($p = 0.0003$), so the portfolio result is not driven by one or two extreme reopening dates.

Two additional tests sharpen the timing interpretation. The weekend prediction-market revision does not simply identify stocks with a persistent tendency to experience opening gaps: the same signal does not predict the immediately preceding opening ($\hat{\beta} = -0.832$, $p = 0.441$) or the following session’s opening ($\hat{\beta} = 2.865$, $p = 0.402$). The predictive relation is therefore concentrated at the first equity reopening associated with the hard-closure signal.

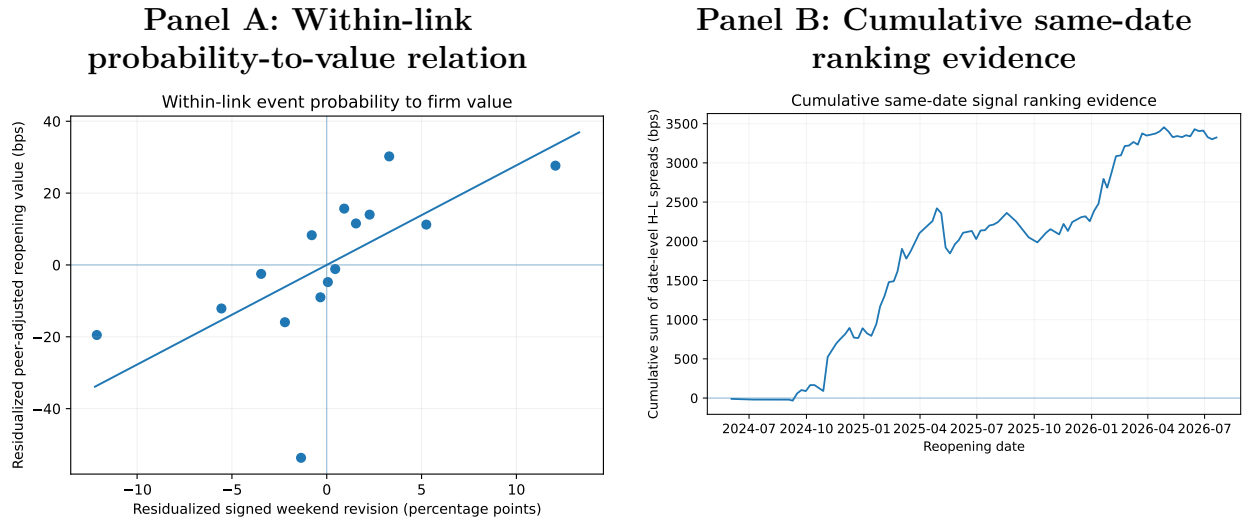


Figure 2. **Event probability to firm value.** Panel A plots the within-link relation between signed prediction-market revisions and peer-adjusted reopening values after removing reviewed-link and reopening-date fixed effects and predetermined controls. Panel B shows the cumulative same-date high-minus-low portfolio spread across reopening dates. The corresponding date-by-date spreads are reported in Appendix Figure A1.

5.1 Closure states: holiday embargoes and the duration null

If longer closures allow more information to accumulate, holiday-extended weekends provide a useful additional test. Restricting the analysis to the twelve holiday-extended reopening dates gives a slope of 5.72 basis points per percentage point on 167 observations — twice the full-sample estimate. Because this subsample contains only twelve reopening dates, conventional clustered inference is less reliable. I therefore also use two small-sample procedures: an exact date-level sign-flip test over all $2^{12} = 4,096$ sign configurations, which asks whether the result survives reversing the direction of date-level shocks, and a 4,999-draw within-date permutation test, which compares the observed signal-stock matching with random assignments on the same date. Both reject the null ($p = 0.0076$ and $p = 0.0196$). The result is also not driven by

a single holiday weekend: every leave-one-date-out slope remains positive, ranging from 4.09 to 7.32.

The holiday result does not imply that the effect increases smoothly with every additional hour of market closure. To test this directly, I interact the weekend signal with total closure duration:

$$Y_{l,d} = \beta_1 S_{l,d} + \beta_2 (S_{l,d} \times \widetilde{H}_d) + \alpha_s + \lambda_d + X'_{s,d-1} \gamma + \varepsilon_{l,d}, \quad (7)$$

where $\widetilde{H}_d$ is standardized total closure duration. The interaction coefficient is $\widehat{\beta}_2 = 1.824$. It is not statistically significant under the clustered or wild-bootstrap procedures ($p = 0.0775$ and $p = 0.0860$), although the within-date permutation test gives $p = 0.0325$. The evidence for a smooth duration effect is therefore suggestive but not robust across inference methods.

Taken together, the results point to a more limited interpretation. Holiday-extended closures are associated with a larger PM-stock relation, but the data do not show that each additional hour of closure mechanically increases the effect. Longer closures may allow more information to accumulate, but holiday weekends can also differ in the type and importance of news that arrives. The holiday estimate should therefore be viewed as supporting evidence from unusually long closures, not as evidence of a constant effect per additional closed hour.

6 The corroboration premium

Having established that the weekend revision predicts the stock move, I next ask which components of that revision are most informative. Prediction markets are still extremely thin when compared with equities, often populated by traders whose motives are recreational, and subject to a plethora of design-dependent biases cataloged by [Wolfers and Zitzewitz \(2004\)](#). A price change on such a venue is an informative but rather noisy signal. Replication could be a natural test of signal quality: does the equity response strengthen when the same economic movement is reproduced by an independent venue or by related contracts? This section provides both tests.

6.1 Replication across venues

Kalshi and Polymarket are very different in their contract designs, fee structures, regulatory statuses, and trader populations. When the same stock-weekend carries a valid economically linked signal on both, agreement between them is a natural corroboration test. On the 480 complete same-stock-weekend observations with a valid signal from each platform, I estimate

$$Y_{s,d} = \beta_1 \bar{S}_{s,d} + \beta_2 A_{s,d} + \beta_3 (\bar{S}_{s,d} \times A_{s,d}) + \alpha_s + \lambda_d + X'_{s,d-1} \gamma + \varepsilon_{s,d}, \quad (8)$$

where $\bar{S}_{s,d}$ is the average of the two platform-level signals and $A_{s,d}$ equals one when the Kalshi and Polymarket signals move in the same economic direction. Within each stock-date-platform cell the platform signal is the largest-absolute valid signed revision, so the test does not require the identical contract to be listed on both venues — an important design detail, since exact contract matching across platforms would shrink the sample significantly.

In 43.5 percent of observations the two platform-level signals agree in economic direction. The corroboration interaction is $\hat{\beta}_3 = 6.443$ with $p = 0.0179$ under two-way clustering by stock and reopening date. The split is very clear: in the agreement subsample the average-signal slope is 6.785 with $p < 0.0001$; in the disagreement subsample it is -0.547 with $p = 0.8395$. When the two venues put forth the same economic signal, the stock response is strong; when they conflict, the estimated slope is indistinguishable from zero.

One alternative reading is that agreement merely proxies for one of the venues being systematically more informative, but the disagreement sample does not support that interpretation. On the 249 disagreement observations from matched cross-platform event groups, entering both platform signals jointly and testing equality of coefficients gives a difference of 1.375 with $p = 0.8603$; neither coefficient is individually significant. Among stock-weekends in which Kalshi and Polymarket disagree, the stock moves in the direction implied by Kalshi in 37.3% of cases and in the direction implied by Polymarket in 46.6% of cases. Although Polymarket has the higher raw success rate, the difference is not statistically significant with $p = 0.1279$. Thus, the disagreement sample provides no evidence that one platform is systematically more informative than the other.

6.2 Replication across contracts

The second corroboration test operates within platforms. Many events are described by several sibling contracts — different thresholds, dates, or phrasings of the same underlying outcome. Event families are determined using the prediction markets event-group metadata: contracts sharing the same named underlying event. Within each stock-date family, the contract with the largest absolute signed revision is the leader; the sibling-common component is the mean signed revision among the other contracts in that family, and the leader-specific residual is the leader revision minus that common component. If prediction-market prices reveal a latent event state rather than a collection of unrelated wagers, then the signal component shared across sibling contracts should carry the information rather than the contract-specific residual.

I enter the sibling-common component and a leader-specific residual jointly in the regression:

$$S_{l,d}^{\text{lead}} = S_{l,d}^{\text{com}} + S_{l,d}^{\text{res}}, \quad Y_{l,d} = \beta_c S_{l,d}^{\text{com}} + \beta_r S_{l,d}^{\text{res}} + \alpha_l + \lambda_d + X'_{s,d-1} \gamma + \varepsilon_{l,d}. \quad (9)$$

A concrete example illustrates the decomposition. On the February 18, 2025 reopening, GEO Group (GEO) was linked to three Kalshi contracts asking how many deportations would occur during Trump’s first year, with thresholds of 250,000, 500,000, and 750,000. All three links carry an economic sign of +1: a higher probability of greater deportation numbers is economically favorable for a detention operator. Table 5 shows how the three contract revisions are converted into the common and leader-specific components.

In this example all three contracts move in the same economic direction, but the leader moves substantially farther. The decomposition assigns -5.5 percentage points to the event-family component supported by the sibling contracts and the remaining -6.5 points to leader specific movement. The empirical question is whether, across all such families, stock prices

Table 5. **Illustrative sibling-family decomposition.** The table reports the signed Friday 8:00 p.m. to Sunday 8:00 p.m. probability revision for each contract in the GEO deportation family associated with the February 18, 2025 reopening. Because the reviewed sign is +1 for all three links, the signed revisions equal the raw probability revisions.

Kalshi contract	Market key	Signed revision	Role
At least 250,000 deportations	KXDEPORTCOUNT-25-250	−3.0	Sibling
At least 500,000 deportations	KXDEPORTCOUNT-25-500	−12.0	Leader
At least 750,000 deportations	KXDEPORTCOUNT-25-750	−8.0	Sibling
Sibling-common component	$(-3.0 + (-8.0))/2$	−5.5	Common
Leader-specific residual	$-12.0 - (-5.5)$	−6.5	Residual

respond more strongly to the former than to the latter.

On 689 observations across 21 stocks and 88 dates, $\hat{\beta}_c = 6.662$ ($p = 0.0084$) and $\hat{\beta}_r = -0.166$ ($p = 0.8382$). The difference $\hat{\beta}_c - \hat{\beta}_r = 6.828$ has $p = 0.0045$, and leave-one-event-family-out common coefficients remain positive, ranging from 5.8 to 13.5. However, a date-cluster wild bootstrap for the common component is weaker ($p = 0.101$), so the finding is not equally strong under every inference procedure. Movement unique to the single contract with the biggest revision — the very contract that the primary specification uses, chosen precisely because it moved most — carries no incremental predictive content once the shared component is extracted.

Taken together, the two corroboration tests show that signal magnitude is not the whole story. Movements that are independently reproduced across venues or across related contracts carry more equity-relevant information than isolated one. Corroboration therefore acts as an information-quality filter, and this principle is potentially useful in any setting with multiple noisy observers of a common latent state — analyst forecasts, alternative data vendors, machine-generated signals, and other settings. Appendix Table A2 reports the corresponding coefficient details for both corroboration exercises.

7 The firm as a portfolio of event exposures

The one-link leading revision design used in the main specification of Section 5 is intentionally transparent, but it is not necessarily the best representation of a firm’s exposure to the world. A large firm can simultaneously be affected by elections, regulation, litigation, tariffs, product approvals, competitor outcomes, merger decisions, and geopolitical states — and prediction markets can price many of these. The aggregation tests in this section ask whether the firm-level event surface contains more information than the single contract alone.

On a common sample of 1,278 stock-weekends, I compare seven ways of combining multiple

linked contracts into a single firm-level event signal. The simplest benchmark keeps only the contract with the largest absolute signed revision. The mean and median use all linked signed revisions with equal weight, while the log-notional-weighted mean gives more weight to contracts with greater weekend trading activity. The top-three capped sum keeps the three largest absolute revisions, caps each at ± 25 percentage points, and adds them. The two remaining measures use a broader set of linked exposures:

$$S_{s,d}^{\text{cap}} = \sum_{m \in \mathcal{M}_{s,d}} \text{clip}(S_{m,d}; -25, +25), \quad S_{s,d}^{\text{fam}} = \sum_{f \in \mathcal{F}_{s,d}} \text{clip}(S_{f^*,d}; -25, +25), \quad (10)$$

where $\mathcal{M}_{s,d}$ is the set of linked contracts for stock s on weekend d , $\mathcal{F}_{s,d}$ is the set of distinct underlying event families, and f^* is the contract with the largest absolute signed revision within family f . The first measure therefore accumulates information across all linked contracts, whereas the second first collapses closely related contracts to a single representative signal and then accumulates across distinct events. In both cases, the ± 25 -point cap limits the influence of an extreme individual contract without eliminating the possibility that several separate exposures jointly matter.

Because these aggregation rules have inherently different scales, their raw regression coefficients are not directly comparable. A sum mechanically becomes larger as the number of linked contracts increases, whereas a mean, median, or single-contract measure remains on approximately the scale of one probability revision. I therefore standardize both the peer-adjusted reopening outcome and each candidate signal on the same common support before estimating otherwise identical stock- and date-fixed-effect specifications. Each coefficient can consequently be read as the change in the reopening outcome, in standard-deviation units, associated with a one-standard-deviation increase in that particular aggregation measure.

Figure 3 shows the pattern and Appendix Table A3 reports the full coefficient set. The event-family capped sum produces the largest standardized coefficient, 0.0806 ($p = 0.0012$), closely followed by the all-contract capped sum at 0.0794 ($p = 0.0019$). The median (0.0702, $p = 0.0301$) and simple mean (0.0694, $p = 0.0445$) are also positive, while the log-notional-weighted mean is similar in magnitude but just outside conventional 5% significance (0.0682, $p = 0.0514$). The top-three capped sum is smaller at 0.0579 ($p = 0.1058$), and the leading-contract rule is weakest at 0.0315 ($p = 0.3607$). Thus, the pattern is not peculiar to a single aggregation formula: measures that retain information from several linked exposures generally outperform the rule that keeps only the largest-moving contract. When the capped sum and simple mean enter jointly, the capped sum retains a coefficient of 0.0780 ($p = 0.0175$) while the mean falls to 0.0027 ($p = 0.953$). The capped-sum coefficient is particularly large among stock-weekends with at least five linked contracts, at 0.142 ($p = 0.00085$). The appropriate conclusion is therefore not that one formula dominates mechanically, but that preserving several distinct event exposures contains substantially more equity-relevant information than selecting a single largest revision.

The aggregation results also address a methodological concern: the design in Section 5 might appear to manufacture the result by selecting whichever contract happened to move most. If that were true, the leading contract would be the strongest predictor. On common

support it is the weakest, while measures that also retain the firm’s other exposures are the strongest. For this reason usage of the leading contract is not generating the main finding of the paper; if anything, it costs power.

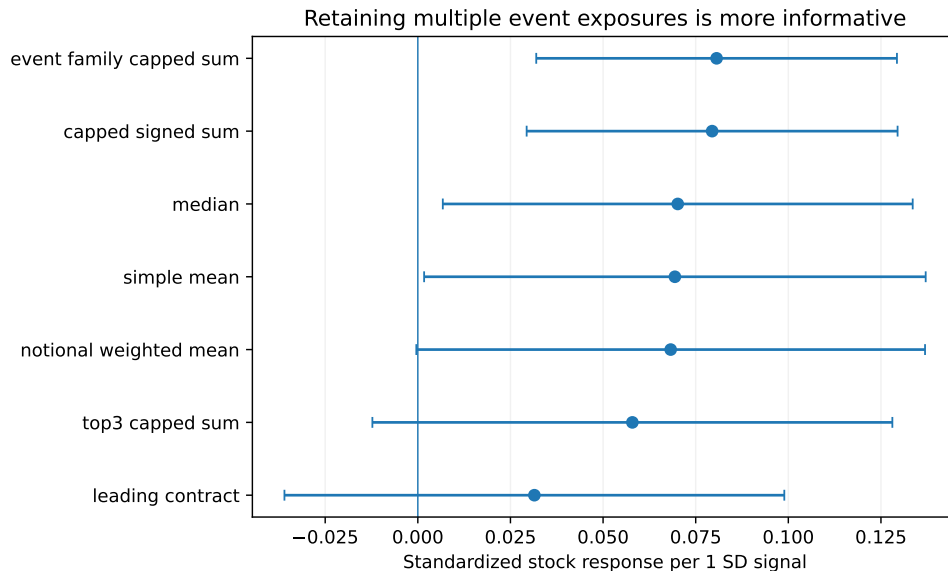


Figure 3. **The firm as a portfolio of event exposures.** Standardized coefficients and 95% intervals compare seven methods of combining a stock-weekend’s linked event revisions on common support. Rules that retain information from several exposures are generally stronger than the leading-contract rule.

The economic interpretation of these results is that a firm’s relationship to the world is a surface of distinct, simultaneously priced states, and that measuring the entire surface beats measuring only its most visible state. These results suggest that prediction-market contracts can be aggregated into a firm-level measure of exposure to real-world events. A conventional factor model writes the firm as a bundle of exposures to broad priced factors. A supply-chain model writes it as a node in a network of customers and suppliers. Textual political-risk measures write it as a bundle of disclosed policy concerns. And the event-market representation shown here sees the firm as a portfolio of explicit probabilistic states. The current paper does not estimate a full event-factor model and does not establish that one aggregation formula is optimal. It does establish that aggregating across distinct economically signed event exposures can contain more equity-relevant information than selecting one dominant contract.

A potentially useful future object is therefore an event beta: the sensitivity of firm value to a set of named event probabilities. The Section 5 coefficient can be read as a local event-probability-to-firm-value exchange rate within repeated links; the aggregation results show that several such rates can coexist at the firm level. This creates a bridge from event studies to a continuous exposure measurement system.

8 Silent repricing

The preceding results raise two natural concerns. First, the relation between weekend prediction-market revisions and equity reopening values could depend on the particular quote–trade consensus estimator used in Section 4.5. Second, large prediction-market moves may simply attract attention and generate more trading at the open, rather than contain information about firm value.

I address the first concern by using outcomes constructed directly from one-minute OHLCV trade bars rather than from the consensus reopening price. These measures do not use the BBO midpoint, the quote-spread screen, the quote–trade agreement rule, or the resulting consensus price. They therefore provide a separate test of whether the weekend signal appears in the stock’s early price path, rather than only in the specific reopening estimator used in the main analysis. The outcomes still come from Databento trade data, so this is a robustness check on the price construction rather than an independent data source. A quote-only alternative would be less useful because opening quotes are more prone to being wide, stale, one-sided, or crossed, whereas trade bars are based on executed prices. Trade data still have their own imperfections, but they provide a cleaner basis for measuring the early price path than relying exclusively on the opening quote channel.

The outcomes in this exercise measure the intensity rather than the direction of reopening price discovery. I therefore use the absolute weekend revision, $|S_{s,d}|$. A favorable 15-percentage-point revision and an unfavorable 15-point revision have opposite directional implications for the stock, but either can generate substantial price adjustment, volatility, or trading. Using signed $S_{s,d}$ would incorrectly allow these two observations to offset one another in a regression whose dependent variable is intrinsically non-directional.

For each outcome I estimate

$$Z_{s,d} = \theta |S_{s,d}| + \delta U_{s,d} + \alpha_s + \lambda_d + X'_{s,d-1} \gamma + \varepsilon_{s,d}, \quad (11)$$

where

$$U_{s,d} = p_{s,d}^F (1 - p_{s,d}^F)$$

is the Bernoulli variance implied by the leading contract’s Friday probability. It is largest at a probability of one-half and approaches zero near zero or one, so it provides a pre-window measure of how unresolved the event was. Including $U_{s,d}$ separates the realized weekend revision, $|S_{s,d}|$, from this pre-existing uncertainty. The coefficient of interest, θ , therefore captures the association between the magnitude of the realized weekend revision and subsequent opening-path intensity conditional on prior event uncertainty.

I consider four outcomes. The first-five-minute high–low range measures the total price interval walked during the opening process. First-thirty-minute realized volatility measures the cumulative variation in consecutive one-minute trade-bar closes. To distinguish price adjustment from trading intensity, I also measure five- and thirty-minute volume relative to each stock’s own recent volume benchmark.

Table 6 separates price-path intensity from trading intensity. Larger absolute weekend revisions predict a wider first-five-minute price range ($\hat{\theta} = 1.537$, $p = 0.0021$) and greater

first-thirty-minute realized volatility (0.767, $p = 0.0056$). By contrast, they do not predict abnormal relative volume over either the first five minutes (0.002, $p = 0.6574$) or the first thirty minutes (-0.001 , $p = 0.8330$).

Table 6. **Silent repricing and activity.** Panel A uses opening-path outcomes constructed directly from one-minute OHLCV bars and therefore does not use the quote–trade consensus reopening level. Panel B asks whether raw prediction-market activity predicts the magnitude of the reopening displacement on the 1,278 observation common support.

Outcome	Estimate	p -value	N
<i>Panel A: Stock opening path, coefficient on S</i>			
Five-minute high–low range	1.537	0.0021	1,312
Thirty-minute realized volatility	0.767	0.0056	1,312
Five-minute relative volume (log)	0.002	0.6574	1,309
Thirty-minute relative volume (log)	-0.001	0.8330	1,310
<i>Panel B: Prediction-market activity</i>			
Weekend PM notional	0.257	0.9132	1,278
Weekend PM trade count	0.447	0.9204	1,278

The first objection is therefore addressed directly. The weekend signal appears in the stock’s early price path even when the particular reopening-price estimator used in the main specification is not used. This does not provide an independent stock-data source; rather, it shows that the empirical relation is not confined to the particular quote–trade consensus statistic.

The attention hypothesis requires a different comparison. If large prediction-market revisions mainly worked by attracting attention and stimulating trading, one would expect the stronger price-path response to be accompanied by unusually high equity volume. It is not. Range and realized volatility increase with the magnitude of the PM revision, while relative volume is essentially unchanged at both horizons. The same distinction appears on the prediction-market side. On the 1,278-observation common support, weekend notional has a coefficient of 0.257 ($p = 0.9132$) and trade count a coefficient of 0.447 ($p = 0.9204$) for the magnitude of the reopening displacement.

These results do not rule out every attention or liquidity channel. They do, however, reject the simpler interpretation in which large PM revisions matter because they merely generate an unusually busy prediction market and an unusually busy stock opening. This pattern is what I call silent repricing. Economically meaningful weekend information is associated with a more intensive opening price path — a wider early range and greater realized volatility — without a detectable increase in aggregate trading volume. The result complements the corroboration evidence of Section 6: information content is associated with the size and independent replication of a probability revision, not with how much trading happened around it.

Since [Barclay et al. \(1990\)](#), trading volume has been a standard observable proxy for information arrival, partly because informed agents must trade so that private information could enter prices. Here the directional weekend signal predicts the reopening move, while

signal magnitude predicts a wider early range and greater realized volatility without detectable abnormal volume. The natural explanation is the nature of the information itself. Unlike private information available to only a subset of traders, the prediction-market state can be observed before the equity reopens. Such public information can therefore be incorporated through changes in quotes and transaction prices without requiring an exceptional increase in the number of shares exchanged. The result complements [Barclay and Hendershott \(2003\)](#), who also show that informative off-hours price discovery need not coincide with high trading volume.

More generally, the result cautions against treating trading volume as a sufficient measure of information arrival. Public, continuously observable state variables can contain price-relevant information even when their incorporation is not accompanied by an unusual surge in trading activity.

9 The catch-up clock

The hard closure identifies which market can move first. A separate question is when the equity market catches up. Since the primary outcome is measured over the first five regular-session minutes, it should not be used to claim that information was visible earlier than that cutoff.

I therefore construct horizon-consistent stock prices at a sequence of checkpoints, using only quote and trade information observable by each checkpoint. A level regression asks whether the prediction-market-implied difference is already present in the stock price by a given checkpoint; an interval-return regression asks whether additional incorporation occurs between checkpoints.

Figure 4 traces the process using quote, trade, and consensus prices. The three measures broadly agree on the timing: the PM-related price difference is visible by late premarket and remains present around the opening, although precision differs across price sources.

The strongest premarket evidence comes from transaction prices. At 9:25 a.m., the trade-price coefficient is 2.681 ($p = 0.0159$), while the trade-price change from 9:00 to 9:25 is 0.836 ($p = 0.0474$). Appendix Table A4 reports the complete interval evidence across quote, trade, and consensus constructions rather than selecting a different price source at different points in time. All three estimates are positive from 9:00 to 9:25 a.m. All three then turn negative between 9:25 and the regular-session open, consistent with some partial correction during the opening transition rather than continued movement in the same direction.

As an additional check, I measure the stock price at fixed points after the open using only information available up to each timestamp. The estimated relation with the weekend signal remains positive through sixty minutes, although the individual estimates are imprecise. The corresponding exact-time interval estimates alternate around zero and are individually insignificant, providing no evidence of a systematic post-open continuation. Taken together, the timing results suggest that most of the adjustment occurs by late premarket trading and the opening process rather than through a gradual drift after the open. The data do not reveal exactly when this adjustment begins, because stock-price coverage is not comparable

throughout the Sunday-night and early-premarket period. Some incorporation may therefore occur before the first checkpoints shown in Figure 4. What the data do show is that little predictable adjustment remains once regular trading is underway.

The absence of a later reversal is also informative. The five-minute-to-close coefficient is essentially zero (0.111, $p = 0.9795$), providing no evidence that the reopening relation is merely a temporary price-pressure effect that subsequently unwinds. Appendix Table A4 provides the complete checkpoint, interval, post-open persistence, and closing-reversal estimates.

Checkpoint levels and post-open persistence

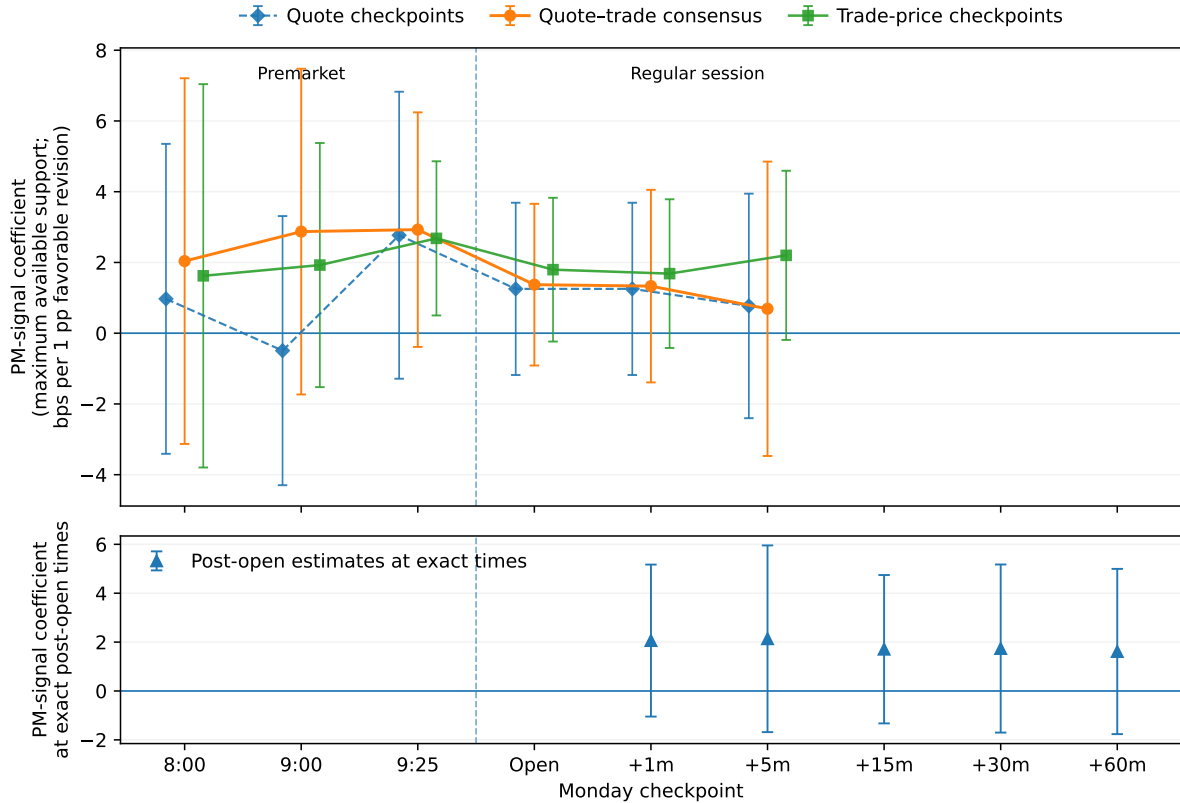


Figure 4. **The catch-up clock.** The upper plot compares quote, trade, and quote-trade consensus estimates across Monday checkpoints. The lower strip reports estimates using stock prices observed at exact post-open times through sixty minutes.

10 Discussion

10.1 The equity market can be closed without being informationally closed

The first implication of the study is that a market can be closed for trading without being closed to information. When the equity market is shut, news relevant to firm value can still be processed elsewhere. In the traditional closure setting, researchers infer this information from what happens when trading resumes, for example through reopening returns or volatility (French and Roll, 1986). Prediction markets make part of that process observable while the stock itself is still unable to trade: the equity price is fixed, but prices of economically related events can continue to change.

This changes what can be learned from nontrading periods. A close-to-open stock return summarizes all information accumulated during the closure in a single price change. Prediction markets can separate part of that information into specific events — an election, a regulatory decision, a court ruling, a geopolitical development, or a product outcome. This makes it possible to ask not only whether relevant news arrived, but what type of event changed, by how much, which firms were exposed to it, and whether the same signal appeared in more than one market.

10.2 Event probabilities as an expectations instrument

A recurring problem in empirical finance is that asset prices respond to surprises, but expectations are usually not observed directly. Researchers therefore rely on market- or data-based proxies that are often specific to a particular setting. Federal funds futures, for example, are used to measure monetary-policy expectations (Bernanke and Kuttner, 2005), while text-based indices capture policy uncertainty (Baker et al., 2016) and analyst forecasts provide expectations about firm-level outcomes.

Prediction markets offer a similar tool for a much wider range of events. They provide continuously updated prices for political, regulatory, legal, geopolitical, product, and other outcomes that are otherwise difficult to quantify. The evidence in this paper shows that changes in these event prices contain information about the value of economically exposed firms, especially when the same movement is corroborated across markets or related contracts. This suggests that prediction-market data could be used to study firm exposure to regulation, litigation, tariffs, elections, geopolitical shocks, supply-chain disruptions, and political connections without constructing a separate expectations measure for each setting.

10.3 The event-probability-to-firm-value exchange rate

The Section 5 coefficient gives a simple economic interpretation to the map between a prediction market and an associated stock. Within a repeated link, β measures how many basis points of linked-stock reopening value correspond to a one percentage point change in the event probability in the favorable direction. It is an event-probability-to-firm-value

exchange rate. The PM probability is noisy, the stock response can combine cash-flow and discount-rate channels, and the relation is local to the observed support. But the coefficient is economically useful precisely because it connects two units that finance usually observes separately: the probability of a real-world state and the market value of a firm exposed to that state.

10.4 Corroboration as an information-quality principle

Prediction markets are only one setting in which many noisy signals about a latent state can be observed. The same problem appears in analyst forecasts, alternative data sources, and in other settings: individual sources may move for idiosyncratic reasons, so agreement across sources can help separate information from noise. That pattern manifests clearly in the paper setting. Equity responses are stronger when Kalshi and Polymarket agree, and within related contract families the common component predicts stock returns while the contract-specific residual does not. By contrast, trading activity itself adds little. The relevant distinction is therefore not between busy and quiet markets, but between isolated and corroborated signals. Agreement does not prove that a signal is correct, but it does identify the part of prediction-market movement that is more likely to contain information relevant for firm value. This principle can be extended beyond prediction markets to any setting in which the same latent state is observed through multiple noisy and partially independent signals.

10.5 Firms as event-exposure portfolios

The aggregation results open a different research program. A firm’s risk can be represented by characteristics, factors, industries, textual topics, network positions, option-implied distributions, and many other approaches. Event markets add another representation: a vector of probabilities for states that affect the firm. At date t , one can imagine an event surface

$$\mathbf{p}_{s,t} = (p_{s1,t}, p_{s2,t}, \dots, p_{sK_s,t}), \quad (12)$$

with an exposure vector

$$\boldsymbol{\beta}_s = (\beta_{s1}, \beta_{s2}, \dots, \beta_{sK_s}). \quad (13)$$

The current paper estimates pieces of this object rather than the full system. But the multi-contract evidence shows why the vector is economically plausible: a bounded combination of several linked state changes contains more information than the single largest contract on common support.

A natural extension is to use these event exposures to build real-time measures of firm risk, estimate event-specific sensitivities across firms, or study how exposure to regulation, geopolitics, and other states changes over time. These are directions for future work. The evidence here is more limited: it shows that combining several economically mapped event probabilities contains information about subsequent equity values.

11 Conclusion

Over roughly forty-eight hours each weekend, US listed equities cannot trade on any venue while markets for the events that determine their value continue to trade without interruption. During those hours, information relevant to firm value continues to be discovered. It is simply discovered somewhere the firm itself is not listed. The probability that a court rules a particular way, that a platform is banned, that a tariff is imposed, or that an election turns is being priced continuously in public markets that do not follow the New York Stock Exchange's calendar, and when the exchange reopens, the stock can adjust toward information that the event market has already incorporated.

That fact matters because of what it implies about the architecture of financial information. For decades, the hierarchy of price discovery has largely been a hierarchy of financial claims: stocks, options and futures on those stocks, indices, and related derivatives, with information moving across markets tied to the same or closely related underlying assets. What the weekend evidence reveals is a layer above that hierarchy — markets not for the assets themselves, but for states of the world that help determine what those assets are worth. In that sense, prediction markets are upstream. They were not designed to serve equity investors, they contain no direct claim on the linked firm, and their participants may have little connection to traditional financial markets. Yet their prices can still move first.

None of this implies that prediction markets are always correct, that their traders systematically outperform professionals, or that the relation documented here can be easily monetized. The paper finds no reliable continuation after the opening process, which suggests that by the time regular equity trading is fully underway, much of the adjustment has already occurred. The narrower conclusion is that markets for events have become part of the process through which information relevant to firm value is discovered, and the weekend closure makes that role unusually visible.

That visibility may not last. Equity trading hours are expanding, and a move toward near-continuous trading would weaken or eventually eliminate the hard closure that makes this setting useful for identification. That is a reason to study the window now rather than a reason to discount it. For the moment, there remains one recurring interval each week in which the world keeps talking and the stock market cannot answer. Prediction markets reveal part of that conversation.

A Additional results and construction diagnostics

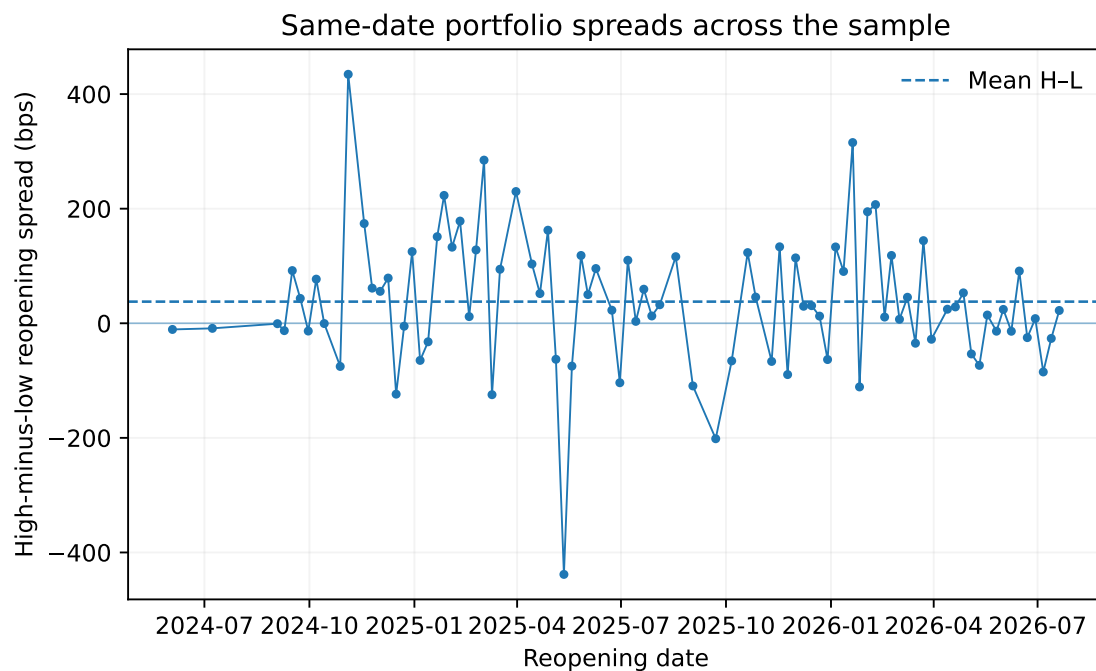


Figure A1. **Same-date high-minus-low spreads by reopening date.** Each point is the date-level high-minus-low peer-adjusted reopening spread; the dashed line marks the sample mean. The figure is an influence diagnostic showing how the result is distributed across reopening dates.

Table A1. **Peer matching, balance, and sensitivity to the peer benchmark.** Panel A compares focal stocks with the mean of their frozen same-date peer sets on the four predetermined matching characteristics. Panel B re-estimates the primary within-link regression under alternative peer-counterfactual constructions while leaving the prediction-market signal, reviewed links, and peer identities unchanged.

Panel A: Post-match balance on primary within-link support			
Characteristic	Focal mean	Peer mean	SMD
Size bucket	2.877	3.025	−0.203
Trailing 20-session gap volatility (bps)	245.23	195.31	0.274
Previous-session intraday return (bps)	−0.80	6.02	−0.027
Previous-session final-30-minute return (bps)	11.59	6.49	0.051
Panel B: Sensitivity to the peer counterfactual			
Specification	$\hat{\beta}$	Stock/date p	Market/date p
Frozen equal-peer benchmark	2.773	0.0281	0.0078
Add all focal-minus-peer matching discrepancies	2.637	0.0326	0.0094
Inverse-distance weighted frozen peers	2.812	0.0227	0.0062
Inverse-distance peers + all mismatch controls	2.708	0.0257	0.0076

SMD denotes standardized mean difference and is reported as a balance diagnostic, not a matching criterion. On the primary within-link support, every focal observation has a peer set; the median contains five peers, the mean is 4.87, and the minimum is three. Inverse-distance weighting keeps the same peer identities and weights each peer by $1/(1 + d)$, where d is the predetermined match distance. **Stock/date and market/date p -values use two-way clustering by stock and reopening date and by prediction market and reopening date, respectively.**

Table A2. **Corroboration coefficient details.** Panel A reports cross-platform corroboration for stock-weekends with valid signals from both venues. Panel B decomposes the leading contract within an event family into movement shared by sibling contracts and a leader-specific residual.

Test	Estimate	S.E.	p-value	N
Panel A: Cross-platform corroboration				
Formal agreement interaction	6.443	2.720	0.0179	480
Slope when platforms agree	6.785	1.699	< 0.0001	208
Slope when platforms disagree	−0.547	2.700	0.8395	272
Panel B: Within-event-family corroboration				
Sibling-common component	6.662	2.527	0.0084	689
Leader-specific residual	−0.166	0.812	0.8382	689
Common minus residual	6.828	2.403	0.0045	689

Table A3. **Alternative aggregation of multiple linked event exposures.** All seven signals are estimated on identical common support. Both the signal and the peer-adjusted reopening outcome are standardized so coefficients are comparable across aggregation rules.

Signal	Construction / rationale	Std. β	Stock/date p	Perm. p
Event-family capped sum	Keep the largest $ S $ within each event family, cap at ± 25 , then sum; limits sibling duplication.	0.0806	0.0012	0.0015
Capped signed sum	Cap every linked S at ± 25 , then sum; retains all exposures while limiting extremes.	0.0794	0.0019	0.0030
Median	Median of linked S ; robust to extreme contracts.	0.0702	0.0301	0.0410
Simple mean	Equal-weight mean of linked S ; uses all links without activity weights.	0.0694	0.0445	0.0345
Log-notional-weighted mean	Weight linked S by log weekend notional; lets more-traded contracts matter more.	0.0682	0.0514	0.0370
Top-three capped sum	Keep the three largest $ S $, cap each at ± 25 , then sum; focuses on the strongest few signals.	0.0579	0.1058	0.0200
Leading contract	Keep only the largest $ S $; transparent single-contract benchmark.	0.0315	0.3607	0.2605

All rows use $N = 1,278$ stock-weekends, 43 stocks, and 91 dates. Stock/date p -values use two-way clustering by stock and reopening date. In a joint capped-sum-plus-mean model, the capped sum remains 0.0780 ($p = 0.0175$) while the mean falls to 0.0027 ($p = 0.953$); a paired-date bootstrap does not reject equality ($p = 0.729$).

Table A4. **Catch-up clock coefficient details.** Panel A reports maximum-support quote, trade, and quote–trade consensus estimates at each Monday checkpoint. Panel B reports the complete set of adjacent-interval estimates for all three price constructions through five minutes after the regular-session open. Panel C reports the separate exact-time post-open estimates used to study persistence after the open. Panel D tests whether the reopening relation reverses by the close.

Panel A: Checkpoint levels

Checkpoint	Quote	Trade	Consensus	N: Q/T/C
8:00 a.m.	0.971 (0.6640)	1.622 (0.5574)	2.038 (0.4397)	288 / 239 / 218
9:00 a.m.	−0.493 (0.7995)	1.926 (0.2740)	2.872 (0.2215)	328 / 355 / 258
9:25 a.m.	2.770 (0.1808)	2.681 (0.0159)	2.928 (0.0835)	344 / 427 / 275
Regular-session open	1.253 (0.3132)	1.796 (0.0832)	1.372 (0.2394)	906 / 1,252 / 831
+1 minute	1.253 (0.3132)	1.684 (0.1164)	1.331 (0.3379)	906 / 1,252 / 839
+5 minutes	0.771 (0.6340)	2.201 (0.0712)	0.690 (0.7452)	1,000 / 1,198 / 855

Panel B: Complete adjacent-interval estimates

Interval	Quote	Trade	Consensus	N: Q/T/C
9:00 → 9:25 a.m.	0.910 (0.1038)	0.836 (0.0474)	0.759 (0.1188)	303 / 317 / 237
9:25 a.m. → open	−1.661 (0.2368)	−0.409 (0.4239)	−1.562 (0.1191)	330 / 428 / 245
Open → +1 minute	0.000 (n/a)	−0.079 (0.7910)	0.110 (0.6384)	919 / 1,269 / 801
+1 → +5 minutes	−0.668 (0.2809)	−0.500 (0.4319)	−0.640 (0.4245)	844 / 1,167 / 680

Panel C: Post-open exact-time persistence

Levels at exact post-open times

Checkpoint	Estimate	p-value	N
+1 minute	2.062	0.1937	501
+5 minutes	2.136	0.2730	501
+15 minutes	1.709	0.2698	501
+30 minutes	1.736	0.3224	501
+60 minutes	1.615	0.3491	501

Returns between successive exact-time checkpoints

Interval	Estimate	p-value	N
+1 → +5 minutes	−0.517	0.5404	789
+5 → +15 minutes	−0.703	0.2122	849
+15 → +30 minutes	0.777	0.1372	878
+30 → +60 minutes	−0.587	0.3772	905

Panel D: Reversal by the close

Return interval	Estimate	p-value	N
+1 minute → close	1.209	0.7753	827
+5 minutes → close	0.111	0.9795	827

Entries in Panels A and B are coefficients with two-sided conventional p -values in parentheses; coefficients are basis points of stock-price response per one-percentage-point favorable prediction-market revision. Quote, trade, and consensus estimates use maximum available support, so sample size varies across sources and checkpoints. Panel B reports all available price constructions for each interval. The zero quote estimate from the regular-session open to +1 minute is mechanical at the available one-minute quote resolution. For the 9:25 a.m.-to-open consensus interval, the date-cluster wild-bootstrap and within-date permutation p -values are 0.038 and 0.005. Panel C uses the exact-time construction consistently; Panel D tests reversal by the close.

References

- Alberto Abadie, Susan Athey, Guido W. Imbens, and Jeffrey M. Wooldridge. When should you adjust standard errors for clustering? *Quarterly Journal of Economics*, 138(1):1–35, 2023.
- Yakov Amihud and Haim Mendelson. Trading mechanisms and stock returns: An empirical investigation. *Journal of Finance*, 42(3):533–553, 1987.
- Kenneth J. Arrow, Robert Forsythe, Michael Gorham, Robert Hahn, Robin Hanson, John O. Ledyard, Saul Levmore, Robert Litan, Paul Milgrom, Forrest D. Nelson, George R. Neumann, Marco Ottaviani, Thomas C. Schelling, Robert J. Shiller, Vernon L. Smith, Erik Snowberg, Cass R. Sunstein, Paul C. Tetlock, Philip E. Tetlock, Hal R. Varian, Justin Wolfers, and Eric Zitzewitz. The promise of prediction markets. *Science*, 320(5878):877–878, 2008.
- Elena Asparouhova, Hendrik Bessembinder, and Ivalina Kalcheva. Liquidity biases in asset pricing tests. *Journal of Financial Economics*, 96(2):215–237, 2010.
- Elena Asparouhova, Hendrik Bessembinder, and Ivalina Kalcheva. Noisy prices and inference regarding returns. *Journal of Finance*, 68(2):665–714, 2013.
- Pavel Atanasov, Phillip Rescober, Eric Stone, Samuel A. Swift, Emile Servan-Schreiber, Philip E. Tetlock, Lyle Ungar, and Barbara Mellers. Distilling the wisdom of crowds: Prediction markets vs. prediction polls. *Management Science*, 63(3):691–706, 2017.
- Scott R. Baker, Nicholas Bloom, and Steven J. Davis. Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4):1593–1636, 2016.
- Michael J. Barclay and Terrence Hendershott. Price discovery and trading after hours. *Review of Financial Studies*, 16(4):1041–1073, 2003.
- Michael J. Barclay and Terrence Hendershott. Liquidity externalities and adverse selection: Evidence from trading after hours. *Journal of Finance*, 59(2):681–710, 2004.
- Michael J. Barclay, Robert H. Litzenberger, and Jerold B. Warner. Private information, trading volume, and stock-return variances. *Review of Financial Studies*, 3(2):233–253, 1990.
- Ben S. Bernanke and Kenneth N. Kuttner. What explains the stock market’s reaction to federal reserve policy? *Journal of Finance*, 60(3):1221–1257, 2005.
- Bruno Biais, Pierre Hillion, and Chester Spatt. Price discovery and learning during the preopening period in the paris bourse. *Journal of Political Economy*, 107(6):1218–1248, 1999.
- Marshall E. Blume and Robert F. Stambaugh. Biases in computed returns: An application to the size effect. *Journal of Financial Economics*, 12(3):387–404, 1983.

- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, 29(2):238–249, 2011.
- Charles Cao, Eric Ghysels, and Frank Hatheway. Price discovery without trading: Evidence from the nasdaq preopening. *Journal of Finance*, 55(3):1339–1365, 2000.
- Sugato Chakravarty, Huseyin Gulen, and Stewart Mayhew. Informed trading in stock and option markets. *Journal of Finance*, 59(3):1235–1257, 2004.
- Lauren Cohen and Andrea Frazzini. Economic links and predictable returns. *Journal of Finance*, 63(4):1977–2011, 2008.
- Arindrajit Dube, Ethan Kaplan, and Suresh Naidu. Coups, corporations, and classified information. *Quarterly Journal of Economics*, 126(3):1375–1409, 2011.
- David Easley, Maureen O’Hara, and P. S. Srinivas. Option volume and stock prices: Evidence on where informed traders trade. *Journal of Finance*, 53(2):431–465, 1998.
- Raymond Fisman. Estimating the value of political connections. *American Economic Review*, 91(4):1095–1102, 2001.
- Robert Forsythe, Forrest Nelson, George R. Neumann, and Jack Wright. Anatomy of an experimental political stock market. *American Economic Review*, 82(5):1142–1161, 1992.
- Thierry Foucault and Albert J. Menkveld. Competition for order flow and smart order routing systems. *Journal of Finance*, 63(1):119–158, 2008.
- Kenneth R. French and Richard Roll. Stock return variances: The arrival of information and the reaction of traders. *Journal of Financial Economics*, 17:5–26, 1986.
- Mason S. Gerety and J. Harold Mulherin. Trading halts and market activity: An analysis of volume at the open and the close. *Journal of Finance*, 47(5):1765–1784, 1992.
- Sanford J. Grossman and Joseph E. Stiglitz. On the impossibility of informationally efficient markets. *American Economic Review*, 70(3):393–408, 1980.
- Joel Hasbrouck. One security, many markets: Determining the contributions to price discovery. *Journal of Finance*, 50(4):1175–1199, 1995.
- Joel Hasbrouck. Intraday price formation in u.s. equity index markets. *Journal of Finance*, 58(6):2375–2400, 2003.
- Tarek A. Hassan, Stephan Hollander, Laurence van Lent, and Ahmed Tahoun. Firm-level political risk: Measurement and effects. *Quarterly Journal of Economics*, 134(4):2135–2202, 2019.
- Friedrich A. Hayek. The use of knowledge in society. *American Economic Review*, 35(4):519–530, 1945.

- Bryan Kelly, L'uboš Pástor, and Pietro Veronesi. The price of political uncertainty: Theory and evidence from the option market. *Journal of Finance*, 71(5):2417–2480, 2016.
- Peter Koudijs. Those who know most: Insider trading in eighteenth-century amsterdam. *Journal of Political Economy*, 123(6):1356–1409, 2015.
- Peter Koudijs. The boats that did not sail: Asset price volatility in a natural experiment. *Journal of Finance*, 71(3):1185–1226, 2016.
- Ananth Madhavan and Venkatesh Panchapagesan. Price discovery in auction markets: A look inside the black box. *Review of Financial Studies*, 13(3):627–658, 2000.
- Dmitriy Muravyev, Neil D. Pearson, and John Paul Broussard. Is there price discovery in equity options? *Journal of Financial Economics*, 107(2):259–283, 2013.
- Maureen O'Hara and Mao Ye. Is market fragmentation harming market quality? *Journal of Financial Economics*, 100(3):459–474, 2011.
- Michael Ostrovsky. Information aggregation in dynamic markets with strategic traders. *Econometrica*, 80(6):2595–2647, 2012.
- L'uboš Pástor and Pietro Veronesi. Uncertainty about government policy and stock prices. *Journal of Finance*, 67(4):1219–1264, 2012.
- L'uboš Pástor and Pietro Veronesi. Political uncertainty and risk premia. *Journal of Financial Economics*, 110(3):520–545, 2013.
- Erik Snowberg, Justin Wolfers, and Eric Zitzewitz. Partisan impacts on the economy: Evidence from prediction markets and close elections. *Quarterly Journal of Economics*, 122(2):807–829, 2007.
- Hans R. Stoll and Robert E. Whaley. Stock market structure and volatility. *Review of Financial Studies*, 3(1):37–71, 1990.
- Alexander F. Wagner, Richard J. Zeckhauser, and Alexandre Ziegler. Company stock price reactions to the 2016 election shock: Trump, taxes, and trade. *Journal of Financial Economics*, 130(2):428–451, 2018.
- Justin Wolfers and Eric Zitzewitz. Prediction markets. *Journal of Economic Perspectives*, 18(2):107–126, 2004.